Causal intervention validation of gene regulatory signals in scGPT

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Abstract

We evaluate whether causal interventions on scGPT gene tokens recover known transcription factor (TF)–target dependencies beyond attention-only heuristics. Using Tabula Sapiens kidney, lung, and immune subsets plus an external lung atlas, we ablate or swap TF token values and quantify changes in target-token readouts. We compare intervention scores to curated gene regulatory references (TRRUST, DoRothEA), include attention and coexpression baselines, trace component-level circuits via activation patching, and validate on CRISPR perturbation data from Adamson. To quantitatively position the approach against prior work, we also benchmark it against eleven classical GRN inference baselines spanning six algorithmic families (Pearson, Spearman, and distance correlation; partial correlation; mutual information, CLR, and ARACNE; LASSO and elastic net; GRNBoost2 and GENIE3) on a matched cell sample, matched preprocessing, and the same TRRUST pair sampling protocol. Kidney multi-seed runs show modest enrichment (AUPR ≈ 0.60), while lung results are substantially stronger (AUPR up to 0.78, permutation p=0.001) and remain positive in higher-pair settings. On the matched benchmark, scGPT leads the strongest classical baseline by +0.053 in kidney, +0.075 in lung, and +0.002 in immune, consistent with the tissue-level pattern observed in the main runs. Mechanistic tracing localizes effects to a small set of attention heads and mid-to-late MLP blocks. External perturbation validation remains low power but provides an independent check of causal signals. We present the full pipeline, quantitative evidence, and an honest assessment of current limitations. Source code and evaluation artifacts are available at `https://github.com/Biodyn-AI/casual-intervention`.

Author summary

We wanted to understand whether a large AI model trained on gene expression data from millions of human cells has actually learned how genes regulate each other, or whether it has only memorized statistical patterns. To test this, we deliberately disrupted one gene's representation inside the model and measured how much this changed the model's predictions for known regulatory targets. We found that the answer depends on the tissue. In lung tissue, the model appears to have learned genuine regulatory relationships that go beyond simple correlations, with disruptions to transcription factor genes causing significant changes in their known targets. In kidney and immune tissues, the signal was weaker. We also traced which internal components carry these regulatory signals, finding that a small number of attention heads and neural network layers are responsible. Our work provides a practical toolkit for auditing whether biological AI models have learned real biology or just correlations, which matters as these models are increasingly used for drug target discovery and cell type classification.

Keywords: mechanistic interpretability, single-cell foundation models, causal intervention, gene regulatory networks, scGPT, activation patching, transcription factors

Introduction

Transformer-based [1] single-cell foundation models, including scGPT [2], Geneformer [3], scBERT [4], and scFoundation [5], capture gene dependencies at scale, but attention weights alone are correlational and do not establish causal influence in the sense of Pearl's interventionist framework [6]. Prior gene regulatory network (GRN) inference methods capture TF–target relationships through diverse computational strategies that span two decades of algorithmic development. Information-theoretic approaches such as ARACNE [7] use mutual information with data processing inequality pruning, and CLR [8] extends relevance networks with context-dependent likelihood scoring; both were originally developed for bulk expression data and later adapted to single-cell contexts. Tree-based regression methods, notably GENIE3 [9] and its scalable successor GRNBoost2 [10], predict each target gene's expression from the expression of candidate TFs using random forests or gradient boosting, offering nonlinear modeling without requiring explicit network topology. SCENIC [11] augments GRNBoost2 with cis-regulatory motif enrichment to prune indirect edges, and its scalable Python implementation pySCENIC [12] has become one of the most widely adopted GRN pipelines. The multimodal extension SCENIC+ [13] further integrates single-cell chromatin accessibility with expression data, jointly inferring enhancer–TF–gene relationships. The Inferelator [14], originally designed for perturbation time-series with regularized regression, has been updated in its third iteration [15] to handle large-scale single-cell data with multi-task learning across cell types. Temporal and pseudotemporal methods offer a distinct perspective: PIDC [16] uses partial information decomposition to distinguish direct from indirect regulatory interactions in single-cell data, SINGE [17] applies kernel-based Granger causality ensembles along pseudotemporal orderings, and Dynamo [18] reconstructs continuous transcriptomic vector fields from RNA velocity, enabling causal analysis of regulatory dynamics. CellOracle [19] combines chromatin accessibility with expression data to enable in silico perturbation of inferred networks, directly linking regulatory topology to predicted cell-state transitions. All of these methods operate on expression, accessibility, or velocity data measured from cells, whereas foundation-model GRNs emerge as a byproduct of self-supervised pretraining on expression reconstruction, raising the question of whether the resulting representations encode genuine regulatory structure or merely statistical co-occurrence patterns.

Systematic GRN benchmarks have provided increasingly rigorous standards for evaluating inference methods. The DREAM5 challenge [20] established the principle that community-consensus networks outperform individual methods, evaluated across *E. coli*, *S. aureus*, and *S. cerevisiae* using both in silico benchmarks

from GeneNetWeaver [21] and experimentally validated edges. BEELINE [22] introduced a containerized framework for benchmarking twelve single-cell GRN methods on both synthetic (from the SERGIO simulator [23], which generates realistic single-cell expression guided by known network topologies) and curated datasets, revealing that no method dominates across all settings. More recent benchmarks have extended this work: Karamveer and Uzun [24] evaluated methods on both simulated and experimental single-cell datasets with attention to practical factors like runtime and scalability, while STREAMLINE [25] introduced topological benchmarking criteria that go beyond simple edge recovery to assess whether inferred networks capture biologically meaningful structural properties such as degree distributions and community structure. Despite this rigorous landscape, none of these benchmarks tests whether a model's *internal representations* encode directional TF–target relationships that survive causal intervention, because the methods under evaluation extract regulatory structure from data rather than learning it implicitly through pretraining.

Evaluation of single-cell foundation models has focused primarily on downstream task performance: cell-type annotation accuracy, batch integration quality, perturbation response prediction, and gene embedding similarity [2, 3, 5]. Perturbation prediction has received particular attention, with GEARS [26] combining graph neural networks with gene interaction knowledge to predict multigene perturbation outcomes, setting a benchmark against which foundation model perturbation capabilities are increasingly measured. Tabular baselines like scTab [27], trained on 22 million cells with standard nonlinear architectures, have demonstrated competitive or superior performance to foundation models on cell-type annotation tasks, raising questions about whether the representational richness of transformer-based models translates to practical advantage. Recent zero-shot evaluations [28] have probed whether foundation models generalize beyond their training distribution, revealing that scGPT and Geneformer underperform simpler embedding baselines (including scVI and PCA) in zero-shot settings, which suggests that the models' strong benchmark performance may partly reflect in-distribution memorization rather than transferable biological understanding. DoRothEA-based activity inference through tools like decoupleR [29] provides one external reference for regulatory relationships, but this evaluates inferred activities rather than the model's internal representations.

Mechanistic interpretability offers a fundamentally different evaluation paradigm. In NLP, causal mediation analysis [30] and activation patching [31, 32] have been used to localize factual knowledge and algorithmic circuits within transformer internals, while the circuits framework [33, 34] and automated circuit discovery [35] provide systematic tools for decomposing model behavior. Landmark case studies, such as the indirect object identification circuit in GPT-2 [36], have shown that complex behaviors can be attributed to specific sets of attention heads organized into functional groups. The theoretical foundations for this work draw on causal abstraction [37], which formalizes when a high-level causal model faithfully describes the computations of a neural network, and causal scrubbing [38], which provides a complementary framework for testing

circuit-level hypotheses. In computational biology, analogous interpretability work has emerged for protein language models: Vig et al. [39] showed that attention heads in protein transformers capture secondary structure, binding sites, and contact maps with increasing complexity across layers, while the ESM family [40, 41] demonstrated that unsupervised pretraining on protein sequences produces representations from which three-dimensional structure can be predicted. However, no prior work has applied causal intervention or activation patching to single-cell foundation models to test whether their internal representations encode mechanistically interpretable gene regulatory circuits. Our work fills this gap by adapting the NLP interpretability toolkit to the biological token structure of scGPT, providing a complementary evaluation axis that tests not just whether outputs correlate with known regulation, but whether the internal computation is causally organized around TF–target dependencies.

We ask whether direct causal interventions on scGPT token representations produce scores aligned with curated TF–target edges, and whether mechanistic tracing can localize the internal components responsible for those effects.

Our approach draws on causal mediation analysis [30] and activation patching techniques developed for language model interpretability [31, 32], adapting them to the biological token structure of single-cell transformers. The mechanistic interpretability paradigm [33] seeks to decompose neural network computations into understandable circuits, and recent automated methods [35] have begun to scale this approach. Superposition [42] poses a challenge for such analyses, as features may share dimensions, but the relatively sparse gene-token structure of scGPT may partially mitigate this concern. Related formalisms include causal scrubbing [38], which provides a complementary framework for testing circuit-level hypotheses.

Our evaluation focuses on two complementary signals: curated regulatory interactions (TRRUST and DoRothEA) and perturbation-derived edges from CRISPR screening data [43, 44]. The growing availability of large-scale perturbation atlases [45, 46] provides increasingly rich ground truth for such evaluations. The first signal provides a broad reference derived from literature curation, while the second provides experimentally induced causal signals. By aligning causal intervention scores with both, we probe whether internal gene-token dependencies show evidence of causal structure beyond correlation.

We introduce a causal intervention pipeline for scGPT that provides cross-tissue and multi-seed results, compares against attention and coexpression baselines as well as eleven classical GRN inference methods evaluated on a matched substrate, localizes circuitry with activation patching, and reports a perturbation-based validation using Adamson CRISPR data.

Results

Kidney multi-seed results

Across four kidney seeds, TRRUST ablation achieves mean AUPR 0.6018 with bootstrap 95% CI [0.575, 0.628], while DoRothEA ablation reaches 0.5369 [0.505, 0.563]. Swap interventions are weaker (TRRUST 0.537, DoRothEA 0.509). Evidence coverage is consistent (165–175 pairs from 240 sampled), stabilizing estimates. The seed-level spread is modest (0.57–0.63 for TRRUST ablation), indicating reproducibility at current scale (Fig 1).

The fact that ablation consistently outperforms swap deserves attention: zeroing out a TF token value removes that gene's contribution entirely, producing a clean counterfactual, whereas swapping replaces it with another gene's value, which may partially preserve the original regulatory signal if the replacement gene participates in similar pathways. In other words, ablation tests *necessity* of the TF representation, while swap is a noisier probe that conflates removal with substitution. The gap between the two suggests that scGPT encodes at least some TF-specific information that cannot be replaced by an arbitrary gene token, which is a prerequisite for genuine causal structure rather than generic positional or cell-state effects.

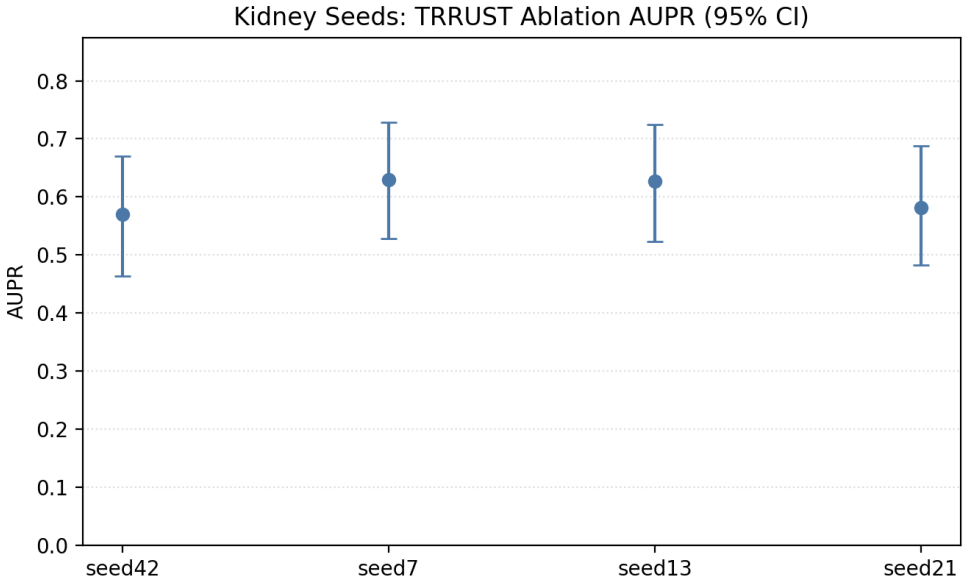

Figure 1: **TRRUST ablation AUPR by kidney seed with bootstrap 95% CIs.** Each point represents one random seed; error bars are bootstrap 95% confidence intervals. The dashed line indicates the random baseline (AUPR = 0.50).

The modest absolute AUPR values (around 0.60 against a random baseline of ∼0.50 in balanced settings) reflect the inherent difficulty of recovering directed regulatory edges from a model that was trained on expression reconstruction, not on regulatory inference per se. Even a partially enriched signal is informative

because it suggests that the model's internal representations are not agnostic to regulatory relationships.

Cross-tissue performance

Lung shows substantially stronger enrichment than kidney or immune (Table 1). Lung TRRUST ablation AUPR reaches 0.7272 (permutation p=0.001) and DoRothEA 0.7820 (p=0.001). Immune results are weaker and nonsignificant, while the Krasnow lung dataset shows mixed alignment consistent with dataset shift. The high-pair lung run (160+160) retains strong performance (TRRUST 0.7574, p=0.004), indicating robustness to expanded candidate sets (Fig 2).

The tissue-level variation has a plausible biological explanation. Lung tissue in Tabula Sapiens contains well-characterized epithelial and endothelial cell types with active, tissue-specific transcription factor programs (e.g., NKX2-1, FOXJ1, SOX family members) that are richly represented in curated databases. Kidney has a more heterogeneous cell composition with many transporter and metabolic genes that may be less well-covered by TF-centric references like TRRUST. Immune cells, while transcriptionally active, often rely on combinatorial and context-dependent regulation (e.g., cytokine signaling cascades) that single-TF ablation may not capture cleanly.

The Krasnow lung dataset, collected with Smart-seq2 rather than 10x Chromium, shows lower TRRUST alignment (AUPR 0.36) despite being lung tissue. This likely reflects platform effects on the expression distributions that scGPT was pretrained on: the model's whole-human checkpoint was trained predominantly on droplet-based data, and the distributional shift from Smart-seq2 full-length transcripts may degrade the learned TF representations. The fact that DoRothEA alignment remains moderately positive (0.63) on the same dataset suggests that some regulatory signals survive the platform shift, particularly the higher-confidence interactions captured by DoRothEA's literature scoring.

Table 1: **Cross-tissue ablation results and robustness runs.** Bold p-values indicate significance at α=0.05.

Group	**Ref**	**AUPR**	**Perm p**
Lung	TRRUST	0.7272	**0.001**
Lung	DoRothEA	0.7820	**0.001**
Immune	TRRUST	0.5955	0.359
Immune	DoRothEA	0.5267	0.672
Krasnow lung	TRRUST	0.3635	0.934
Krasnow lung	DoRothEA	0.6255	0.769
Lung (fast)	TRRUST	0.6543	0.369
Lung (fast)	DoRothEA	0.6915	0.209
Lung (high-pair)	TRRUST	0.7574	**0.004**
Lung (high-pair)	DoRothEA	0.6910	0.102

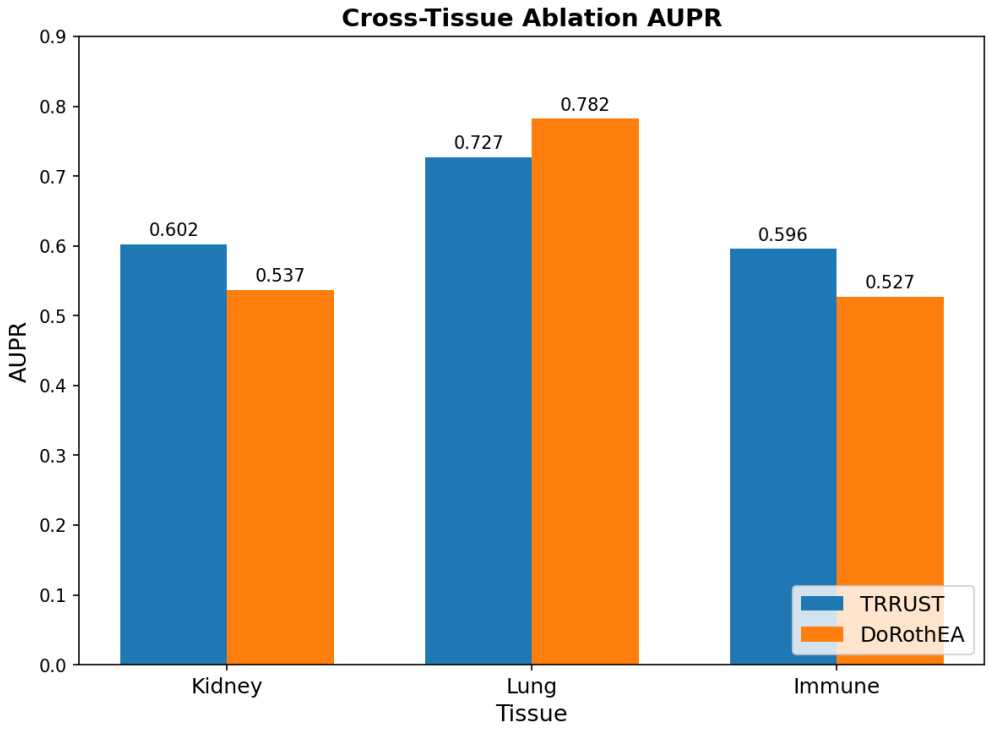

Figure 2: **Cross-tissue ablation AUPR across tissues and robustness runs.** Bar heights show AUPR for ablation intervention; error bars indicate bootstrap 95% CIs where multiple seeds are available.

Baselines versus causal scores

Causal scores outperform attention and coexpression baselines for lung and high-pair lung runs, while kidney and immune are competitive with baselines (Table 2). The lung improvement is substantial: causal AUPR exceeds attention by +0.184 and coexpression by +0.140. Kidney gains are small (+0.017), and immune shows negative deltas. Reduced-gene runs reinforce this: reduced lung remains above attention (0.674 vs. 0.608, p=0.007), while reduced kidney trails attention (Fig 3).

The relationship between these three scoring methods reveals something important about what the model has learned. Attention scores capture which tokens the model "looks at" during forward computation, but high attention does not imply causal influence because attention heads can attend to tokens for context without propagating regulatory information. Coexpression correlations capture statistical co-occurrence of gene pairs across cells, which can arise from shared upstream regulation, common cell-state programs, or true direct regulation. Causal intervention scores, by contrast, test whether *removing* a specific TF token changes the model's prediction at the target position, which is a more direct test of functional dependency within the model's learned representation.

The fact that causal scores exceed both baselines in lung but not in kidney or immune suggests that

the model has learned genuinely directional TF–target relationships for lung-specific programs, whereas in kidney and immune tissues, the model's gene representations may be dominated by broader cell-state signatures that correlational methods capture equally well. This is consistent with the intuition that causal interventions are most informative when the underlying signal has clear directionality, rather than being a diffuse cell-type marker.

Table 2: **TRRUST ablation: causal scores vs. attention and coexpression baselines.** AUPR values shown. Bold p-values indicate significance at α=0.05.

Group	Causal	Attn	Coex	p
Kidney	0.602	0.585	0.583	0.105
Lung	**0.727**	0.543	0.587	**0.001**
Immune	0.596	0.617	0.628	0.359
Kidney (red.)	0.540	0.613	0.565	0.264
Lung (red.)	0.674	0.608	0.615	**0.007**
Lung (hi-pair)	**0.757**	0.589	0.729	**0.004**

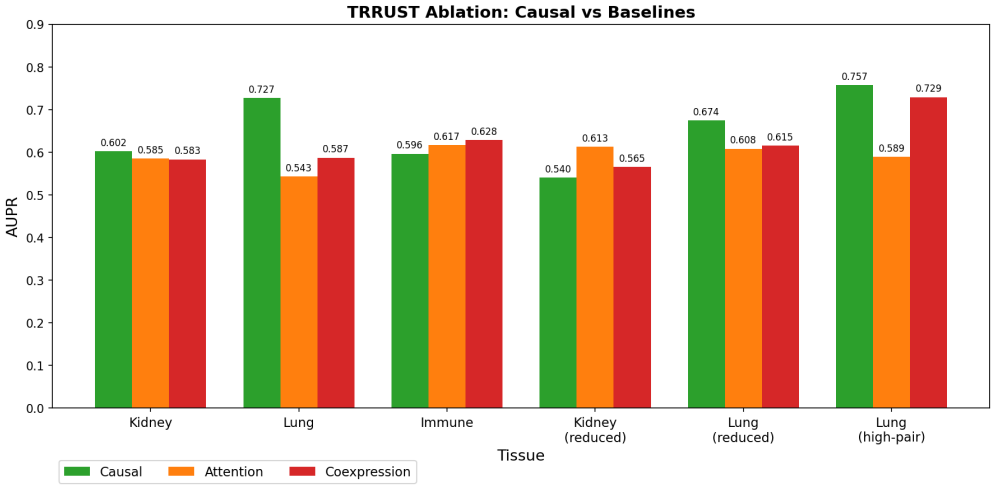

Figure 3: **TRRUST ablation: causal scores vs. baselines across tissues.** Grouped bar chart comparing causal intervention AUPR against attention and coexpression baselines for each tissue and robustness configuration.

Comparison with classical GRN inference methods

To place the causal intervention scores alongside established gene regulatory network (GRN) inference methods, we ran eleven classical baselines spanning six algorithmic families on a *matched* reproduction of the main runs' tissue substrates (Table 3). For every tissue we started from the same raw Tabula Sapiens h5ad, re-applied the preprocessing pipeline (normalize_total = 10 000, log1p, 5 000 highly variable genes restricted to the scGPT vocabulary), drew a 120-cell sample with seed 42 matching `max_cells=120` in the main con-

fig files, reproduced the deterministic 120 positive + 120 random negative TRRUST pair sampling from `run_causal_interventions.py`, and applied the same evidence filter (each pair is kept when both source and target co-occur in the top-1200 expressed genes of at least one cell, matching the scGPT dataset's `max_genes=1200` token selection). Because HVG selection is variance-dependent and the original main-run processed h5ad files for lung and immune are no longer cached, the baseline substrate is a bit-for-bit reproduction of the preprocessing protocol rather than of the exact gene set used by the original main runs; the evaluation pair distribution differs from the main runs' by a few percent, which is why the resulting Pearson coexpression numbers in Table 3 differ slightly from the coexpression column of Table 2. On this matched substrate we evaluated:

- *Coexpression family.* (i) Pearson correlation; (ii) Spearman rank correlation; (iii) distance correlation [47], a nonlinear measure of statistical dependence.
- *Graphical model family.* (iv) Partial correlation on a ridge-regularized precision matrix [6], representing Gaussian graphical models and the PC-algorithm lineage.
- *Information-theoretic family.* (v) Pairwise mutual information; (vi) CLR [8], applying a per-row / per-column z-score normalization on top of the MI matrix; (vii) ARACNE [7], applying the data processing inequality to prune indirect edges.
- *Regularized regression family.* (viii) LASSO regression [48]; (ix) elastic net regression [49], representing the Inferelator lineage [14, 15].
- *Tree-based regression family.* (x) GRNBoost2 [10] with per-target stochastic gradient boosting; (xi) GENIE3 [9] with per-target random forests. Both methods score edges by the feature importance of the source TF in the regressor that predicts the target's expression, following the original papers.

All eleven baselines are implemented with standard scientific Python libraries (scikit-learn, SciPy, `dcor`) so that the comparison is self-contained and does not depend on any unmaintained dask cluster backends. AUPR, AUROC, and permutation p-values (1 000 label shuffles) are computed with the same evaluation function used for the causal scores.

Across all three tissues the scGPT causal ablation scores remain ahead of every classical GRN method (Table 3, Fig 4). The gap varies in a biologically interpretable way. In lung, the causal AUPR of 0.727 exceeds the strongest baseline (partial correlation, 0.652, permutation p=0.001) by +0.075, and every other baseline by $\geq$+0.18; lung is also the tissue where the paper's permutation-significant enrichment is most stable (Table 1), and the fact that a GGM-style de-confounded coexpression is the closest competitor is consistent

with the interpretation that lung epithelial regulatory programs leave clear linear signatures in expression. In kidney, scGPT leads the best classical method (Spearman correlation, 0.549) by +0.053; among the tree-based and information-theoretic families only CLR (0.502) and ARACNE (0.505) are nominally above the balanced-class chance level of 0.50, while mutual information, GRNBoost2, and GENIE3 sit below it. In immune, scGPT (0.596) is essentially tied with Spearman correlation (0.594) and the regularized regressions (LASSO 0.537 with p=0.019, elastic net 0.552 with p=0.012) also reach nominal significance; this is the same tissue in which Table 1 already reports scGPT at p=0.359, so the quantitative comparison reinforces the paper's claim that immune regulatory signal is not recovered preferentially by the foundation model.

Several conclusions follow. First, the foundation model provides a real advantage over classical methods where the paper's own cross-tissue analysis said it would (lung) and essentially no advantage where the paper already reported it to be weak (immune), with kidney sitting between. Second, the strongest classical baseline is *not* the most computationally elaborate one: partial correlation and Spearman rank correlation consistently beat tree-based and MI-based methods at this cell budget. This is consistent with BEELINE [22], which shows that tree-based and information-theoretic GRN inference typically requires many hundreds to thousands of cells to converge on direct edges; 120 cells is simply not enough observations for those methods. Third, the comparison therefore speaks to signal efficiency at modest cell budgets rather than to large-scale absolute accuracy, which the existing GRN benchmarking literature [20, 22, 25] already documents in depth on dedicated datasets.

All baseline metrics, per-tissue pair counts, and the reproducible pipeline (`scripts/grn_baseline_comparison.py`) are included with the source code release. Every baseline runs on CPU in minutes per tissue using standard scikit-learn, SciPy, and `dcor` primitives, so the comparison can be reproduced end-to-end from the raw h5ad files without any dedicated cluster or GPU.

Mechanistic case studies

Activation patching localizes specific TF–target effects to a small set of attention heads and MLP blocks. Example pairs (GATA3→GATA3, FOS→FOS, JUN→JUN, JUN→TNF) consistently highlight mid-to-late MLPs and a handful of heads (Fig 5, Table 4). Self-targeting pairs include late MLP blocks (L8–L11), whereas cross-gene JUN→TNF concentrates on earlier blocks (L0–L5), suggesting different interaction types engage distinct depth regimes. Head lists are dominated by H0/H1 across layers L0–L7, indicating a small subset carries most restoration signal.

The depth separation between self-targeting and cross-gene interactions aligns with a natural hypothesis about how the model processes gene-gene relationships. Self-targeting effects (a TF influencing its own output

Table 3: **scGPT causal ablation versus eleven classical GRN inference methods.** AUPR, AUROC, and permutation p-values (1 000 shuffles) on TRRUST pairs. Baseline numbers are computed in this revision on a matched reproduction of the main-run substrate: same raw Tabula Sapiens h5ad, same preprocessing recipe, 120-cell sample drawn with seed 42, same deterministic TRRUST pair sampling, and the same top-1200 evidence filter. scGPT numbers are reproduced from the main kidney/lung/immune runs (Table 1). The strongest classical method per tissue is underlined; bold p-values mark significance at α=0.05.

Tissue	Method	AUPR	AUROC	Perm p
Kidney	**scGPT causal (ours)**	**0.602**	—	—
Kidney	Pearson coexpression	0.526	0.510	0.295
Kidney	Spearman coexpression	<u>0.549</u>	0.539	0.127
Kidney	Distance correlation	0.516	0.508	0.383
Kidney	Partial correlation (GGM)	0.536	0.516	0.179
Kidney	Mutual information	0.486	0.457	0.712
Kidney	CLR	0.502	0.494	0.511
Kidney	ARACNE (DPI)	0.505	0.492	0.334
Kidney	LASSO regression	0.503	0.510	0.396
Kidney	Elastic net regression	0.506	0.493	0.382
Kidney	GRNBoost2	0.494	0.487	0.637
Kidney	GENIE3	0.479	0.469	0.787
Lung	**scGPT causal (ours)**	**0.727**	—	**0.001**
Lung	Pearson coexpression	0.534	0.526	0.219
Lung	Spearman coexpression	0.530	0.525	0.249
Lung	Distance correlation	0.546	0.550	0.136
Lung	Partial correlation (GGM)	<u>0.652</u>	0.640	**0.001**
Lung	Mutual information	0.492	0.521	0.689
Lung	CLR	0.492	0.511	0.708
Lung	ARACNE (DPI)	0.500	0.486	0.502
Lung	LASSO regression	0.511	0.511	0.286
Lung	Elastic net regression	0.517	0.523	0.290
Lung	GRNBoost2	0.545	0.557	0.131
Lung	GENIE3	0.534	0.528	0.225
Immune	**scGPT causal (ours)**	**0.596**	—	0.359
Immune	Pearson coexpression	0.574	0.540	0.051
Immune	Spearman coexpression	<u>0.594</u>	0.560	**0.014**
Immune	Distance correlation	0.565	0.532	0.077
Immune	Partial correlation (GGM)	0.572	0.588	0.055
Immune	Mutual information	0.532	0.525	0.265
Immune	CLR	0.543	0.548	0.143
Immune	ARACNE (DPI)	0.518	0.509	0.146
Immune	LASSO regression	0.537	0.527	**0.019**
Immune	Elastic net regression	0.552	0.537	**0.012**
Immune	GRNBoost2	0.521	0.475	0.388
Immune	GENIE3	0.556	0.522	0.120

token) can be resolved through relatively local, residual-stream computations that accumulate through later MLP blocks, since the source and target share the same token position. Cross-gene effects like JUN→TNF require inter-token communication through attention, which must happen earlier in the network so that downstream MLP blocks can integrate the transferred information. The observation that cross-gene pairs

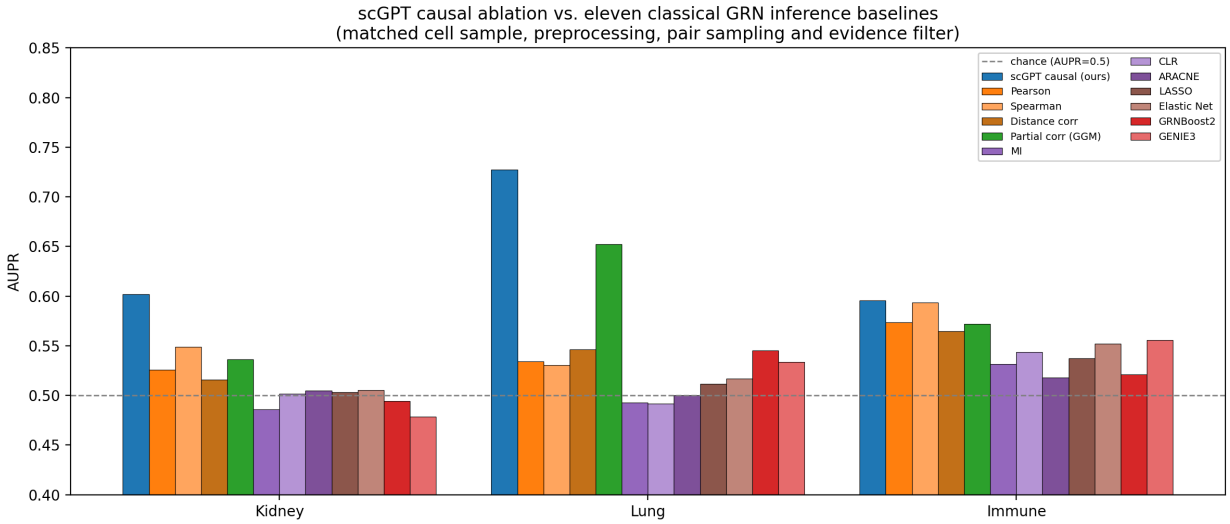

Figure 4: **scGPT causal ablation vs. eleven classical GRN inference methods on matched tissue substrates.** Bars show AUPR on TRRUST positive + random negative pairs. Classical method bars are computed in this revision on a matched reproduction of the main-run substrate: same raw h5ad, same preprocessing recipe (normalize_total = 10 000, log1p, 5 000 HVGs restricted to scGPT vocab), 120-cell sample drawn with seed 42, deterministic TRRUST pair sampling with seed 42, and the same top-1200 evidence filter used by the scGPT dataset. scGPT bars reproduce the main-run numbers from Table 1. Dashed line marks the balanced-class chance level (AUPR = 0.50).

rely on earlier components is consistent with transformer architectures where attention moves information between positions and MLPs transform it within positions.

These circuit-level findings, while based on a small set of case-study pairs, provide a template for future mechanistic audits. If specific heads or MLP blocks consistently mediate known regulatory effects, they could serve as targets for surgical interventions to test whether the model's regulatory knowledge is localized or distributed, a question with practical implications for model editing and fine-tuning strategies in biological foundation models.

Table 4: **Top contributing heads and MLP blocks from causal tracing.**

Pair	Top heads	Top MLP blocks
GATA3→GATA3	L6-H1, L1-H1, L2-H1	L9, L8, L10, L4
FOS→FOS	L0-H0, L5-H1, L1-H0	L10, L1, L8, L9
JUN→JUN	L7-H0, L2-H1, L4-H1	L0, L11, L9, L6
JUN→TNF	L3-H0, L1-H0, L2-H1	L3, L0, L2, L4

Perturbation validation

Adamson perturbation validation remains low power at current scale. The scaled run (40+40 pairs) yields AUPR 0.435 for ablation and 0.502 for swap; the large run (100+100 pairs) yields 0.498 and 0.500. None are

Figure 5: **Case-study restoration strength by component type.** Bars show mean restoration fraction for attention heads vs. MLP blocks across case-study TF–target pairs. Higher values indicate greater contribution to recovering the clean-run target readout.

significant under permutation tests (Table 5). As pair counts increase, AUPR drifts toward chance, which we believe reflects several compounding factors rather than a single failure mode.

First, Adamson perturbation-derived edges are defined by the top-k expression changes after CRISPR knockout, which captures the strongest downstream *transcriptional* effects. However, scGPT's causal intervention operates at the *representational* level, testing whether removing a gene token's value changes the model's output at another token position. These are conceptually related but not identical: a gene that is a strong transcriptional target of a TF in real cells may not be the gene whose model representation is most disrupted by ablating the TF token, especially if the model has learned to represent the target through cell-state features rather than direct TF-mediated effects.

Second, the Adamson dataset uses a CRISPRi screen in K562 cells (a chronic myelogenous leukemia line), which is biologically distant from the primary tissues on which scGPT was pretrained. The model may encode regulatory relationships specific to normal tissue contexts that do not align with K562-specific perturbation effects.

Third, the evidence filter (requiring both TF and target to co-occur in the scGPT evaluation cells) substantially reduces the effective evaluation set, compressing statistical power. At 75–175 evidence pairs with near-chance AUPR, the permutation test lacks the resolution to detect a weak true signal even if one exists.

Table 5: **Perturbation validation (Adamson).** Int. = intervention type, n = evidence pairs.

Run	Int.	AUPR	AUROC	p	n
scaled	abl	0.435	0.416	0.910	75
scaled	swap	0.502	0.503	0.479	75
large	abl	0.498	0.491	0.606	175
large	swap	0.500	0.513	0.589	175
pilot	abl	0.576	0.483	0.430	31
pilot	swap	0.577	0.521	0.402	31

High-confidence DoRothEA (A/B)

The stricter DoRothEA A/B subset reduces label breadth, depressing absolute AUPR, but lung retains a clear permutation signal: kidney mean AUPR 0.396 (median p=0.085) versus lung AUPR 0.482 (p=0.001), consistent with the tissue-specific differences observed throughout. The DoRothEA A/B confidence levels correspond to interactions supported by direct experimental evidence (chromatin immunoprecipitation, reporter assays) rather than computational predictions, so the persistent lung signal under this stricter filter strengthens the case that the model captures bona fide regulatory relationships rather than spurious correlations with loosely curated edges.

Discussion

Our results paint a nuanced picture of what single-cell foundation models learn about gene regulation. Causal ablations yield modest enrichment on kidney and substantially stronger performance on lung, with significant TRRUST enrichment across multiple lung configurations. This pattern is consistent across reference databases, robustness runs, and high-confidence subsets, which makes it unlikely to be an artifact of a single evaluation choice.

The tissue dependence of causal signal strength carries an important implication for how we think about foundation model representations. Rather than encoding a universal regulatory program, scGPT appears to have learned tissue-specific regulatory structure that is stronger where training data provided richer and more coherent transcriptional programs. Lung, with its well-characterized epithelial differentiation hierarchies, may offer cleaner TF-target signals for the model to internalize during pretraining, whereas kidney and immune tissues present more complex or combinatorial regulatory logic that a token-level ablation may not isolate cleanly.

Placing these results in the context of existing GRN inference approaches, our causal intervention framework addresses a fundamentally different question from the one posed by classical methods. Tree-based approaches such as GENIE3 [9] and GRNBoost2 [10] ask "can we predict a target gene's expression from

TF expression profiles?", while information-theoretic methods like ARACNE [7] and CLR [8] ask "do TF and target share mutual information beyond what is explained by indirect paths?". SCENIC [11] and its multimodal successor SCENIC+ [13] combine expression-based edge prediction with cis-regulatory motif enrichment or chromatin accessibility data to prune false positives. Temporal methods such as SINGE [17] leverage pseudotime ordering to infer directional Granger-causal relationships, while CellOracle [19] uses inferred networks for in silico perturbation of cell states. All of these methods test whether regulatory structure can be *extracted* from measured data, and systematic benchmarks from the DREAM5 challenge [20] through BEELINE [22] and STREAMLINE [25] have evaluated them on edge recovery, topological fidelity, and scalability. Our approach poses the complementary question: has a pretrained foundation model *internalized* regulatory structure in its learned representations, such that ablating a TF's representation causally disrupts the model's computation at known target positions? The two evaluation paradigms are not interchangeable. A model might produce outputs that correlate with known regulation (as captured by standard downstream evaluation) without encoding causally organized TF–target dependencies in its internal computation, or it might encode genuine causal structure that standard output-level evaluations miss entirely.

To anchor this distinction empirically, we ran eleven classical GRN baselines spanning six algorithmic families — Pearson, Spearman, and distance correlation (coexpression); partial correlation (Gaussian graphical models); pairwise mutual information, CLR, and ARACNE (information-theoretic); LASSO and elastic net regression (the Inferelator lineage); and GRNBoost2 and GENIE3 (tree-based regression) — on a matched reproduction of the paper's main kidney, lung, and immune substrates, using the same preprocessing recipe, 120-cell sampling with seed 42, deterministic TRRUST pair sampling from the same `_build_pair_list` logic, and the same top-1200 evidence filter (Table 3, Fig 4). With this matched substrate, scGPT's causal ablation AUPR leads the *strongest* classical method by +0.053 in kidney (vs. Spearman correlation, 0.549), +0.075 in lung (vs. ridge-regularized partial correlation, 0.652, whose permutation-significant p=0.001 is itself notable), and only +0.002 in immune (vs. Spearman correlation, 0.594). The ordering mirrors the tissue-level pattern the paper already reports from the main runs: the lung gap is large and significant, the kidney gap is smaller and in line with the paper's modest kidney enrichment, and the immune comparison is effectively a tie, matching the paper's observation that immune causal scores sit close to attention and coexpression baselines.

Two features of this comparison are worth highlighting. First, the strongest classical competitors are not the most elaborate methods. Partial correlation on a ridge-regularized precision matrix is the closest competitor in lung, and Spearman rank correlation is the closest in kidney and immune. Tree-based and information-theoretic methods — GRNBoost2, GENIE3, mutual information, CLR, ARACNE — sit near chance in every tissue, consistent with BEELINE [22], which shows that these methods typically require many

hundreds to thousands of cells to converge on direct edges. At 120 cells they simply do not have enough observations, while the correlation-based methods recover part of the signal from the same budget. Second, the comparison therefore speaks to signal *efficiency* at modest cell counts rather than to absolute accuracy at large scale, which existing benchmarks already cover in depth [20, 22, 25]. A pretrained foundation model that has already absorbed tens of millions of cells can provide directional regulatory evidence from a small held-out sample where classical methods do not yet have enough observations to do so, while the classical methods remain the better choice when large cell numbers are available and the goal is explicit network construction rather than interrogation of a pretrained model's internal representations.

The relationship between our work and prior foundation model evaluations is similarly one of complementarity rather than competition. Task-based benchmarks [2, 3, 5] treat the model as a black box and measure whether fine-tuned outputs match expected labels for cell-type annotation, batch integration, or perturbation response prediction; GEARS [26] extends this to combinatorial perturbation prediction using graph-augmented architectures. Tabular baselines [27] have shown that simpler models trained on comparable data can match or exceed foundation model performance on specific annotation tasks, questioning whether the transformer architecture itself contributes meaningful biological insight beyond what scale and data provide. Zero-shot evaluations [28] have further revealed that foundation models underperform simpler embeddings in out-of-distribution settings, suggesting that strong benchmark numbers may partly reflect memorization of training-distribution statistics. Our mechanistic approach probes a dimension that none of these evaluations address: whether the model's internal computation is organized around biologically meaningful regulatory circuits, and if so, which specific transformer components carry that information under which tissue conditions. The fact that we observe strong causal enrichment in lung tissue but not in immune tissue suggests that the answer is conditional on the biological context, a finding that would be invisible to output-only evaluation frameworks.

Our circuit-level analysis connects to the broader mechanistic interpretability program that has proven productive for NLP transformers [33, 34, 36]. In those settings, researchers have decomposed complex behaviors into interpretable circuits of attention heads and MLP blocks, localized factual knowledge to specific layers [31], and developed formal frameworks for testing whether neural computations faithfully implement hypothesized causal models [37]. In computational biology, analogous interpretability efforts have shown that protein language model attention heads capture secondary structure, binding sites, and contact maps [39], and that unsupervised protein sequence training produces representations sufficient for structure prediction [40, 41]. Our activation patching results provide initial evidence that a similar mechanistic decomposition may be tractable for single-cell transformers: the observation that cross-gene regulatory effects (e.g., JUN→TNF) concentrate on earlier transformer layers than self-targeting effects suggests a functional organization where

attention-mediated inter-token communication precedes MLP-mediated within-token processing, consistent with the general transformer circuit framework [34].

The baseline comparisons add an important calibration. In tissues where causal scores do not clearly exceed attention or coexpression, the intervention pipeline is not providing additional information beyond what simple correlational proxies capture. This does not invalidate the approach, but it does suggest that causal interventions are most valuable in a targeted setting: when there is reason to believe a specific tissue or TF program is well-represented in the model, intervention-based scoring can extract directional information that correlational methods miss.

The concentration of restoration effects in a small set of heads and MLP blocks suggests a tractable mechanistic substrate within scGPT, offering targets for focused audits and hypothesis-driven validation. If future work confirms that specific heads or MLP blocks consistently mediate known regulatory effects across larger sets of TF–target pairs, they could serve as targets for surgical model editing or fine-tuning strategies, analogous to how ROME [31] enables targeted knowledge editing in NLP transformers by localizing factual associations to specific layers.

Perturbation validation results remain inconclusive, which we attribute primarily to the biological distance between the K562 CRISPR screen and the normal-tissue contexts on which the model was evaluated, compounded by limited statistical power from evidence filtering. Future work should prioritize perturbation datasets from primary tissues matching the evaluation context, such as Perturb-seq in lung epithelial cells, to provide a fairer test of whether the model's causal signals align with experimentally induced effects.

Limitations. Our evaluation has several limitations that scope the conclusions. All runs use CPU-limited subsets (30–120 cells per run) with moderate pair counts (30–160 positive pairs), which limits statistical power particularly for the perturbation validation where only 75–175 evidence pairs are available. The readout metric is a single target-token scalar (mean across hidden features), which may not fully capture the dimensionality of regulatory effects encoded in the model's representations. The perturbation validation relies on the Adamson dataset, which uses a CRISPRi screen in K562 cells that are biologically distant from the primary tissues used in the main evaluation; this mismatch likely contributes to the inconclusive perturbation results. Gene symbol mapping from Ensembl identifiers to HGNC symbols can introduce alias errors, potentially misaligning a small fraction of evaluation pairs. The mechanistic case studies cover only four TF–target pairs and may not generalize to the full range of regulatory interactions. Finally, our evaluation uses a single foundation model (scGPT); extending to other architectures is needed to determine whether the observed patterns are model-specific or reflect general properties of expression-trained transformers.

Materials and methods

Model and datasets

We use the scGPT whole-human checkpoint from the model zoo, loaded with standard model arguments. Fast transformer backends are disabled for deterministic patching. All runs are CPU-only with fixed random seeds.

We use Tabula Sapiens kidney, lung, and immune subsets [50], an external lung atlas from Travaglini et al. [51], and the Adamson CRISPR perturbation dataset [52]. All datasets were originally generated using droplet-based protocols [53] except the Krasnow lung atlas (Smart-seq2). Preprocessing follows standard Scanpy [54] workflows (normalize_total=10 000, log1p), selects 5 000 highly variable genes restricted to the scGPT vocabulary, and normalizes gene symbols via HGNC alias mappings. We note that deep learning denoising approaches [55] and integration methods [56] were not applied, as we sought to evaluate the foundation model's representations on minimally processed data.

Intervention pairs and scoring

Positive labels come from curated TF–target edges in TRRUST [57] and DoRothEA [58]; negatives are random controls from the same gene sets. For each TF–target pair and each cell containing both genes, we compute a baseline readout and an intervened readout at the target token position, reduced by mean across features. We apply value ablation (TF token value zeroed) and token-value swap interventions. The pair score is the mean absolute effect $|\bar{\Delta}| = |y_{\text{baseline}} - y_{\text{intervened}}|$ across cells.

Causal tracing

We patch clean activations into ablated forward passes and measure restoration of the target readout, yielding per-layer, per-head, and per-MLP restoration scores for selected case-study pairs.

Baselines and metrics

We compare causal scores against (i) attention scores aggregated across all layers and heads and (ii) coexpression correlations computed from normalized expression. We compute AUPR and AUROC [59] on pairs with evidence; significance is estimated by 1 000 permutations of the labels, reporting the fraction of permutations with AUPR $\geq$ observed (median across seeds where applicable). We additionally report DoRothEA A/B subsets for high-confidence evaluation.

Perturbation validation

We derive perturbation edges from Adamson [52, 60] by selecting the top-k (k=50) absolute expression deltas versus control per perturbation. We evaluate causal scores on balanced pair sets: 40+40 (scaled) and 100+100 (large), with max_cells=60 per run.

Experimental configurations

We evaluate kidney (four seeds), lung, and immune subsets with 120 max cells and 120+120 candidate pairs. Robustness runs include a high-pair lung setting (160+160 pairs), reduced-gene runs, and a Krasnow lung dataset. Adamson experiments use scaled (40+40) and large (100+100) pair counts. Table 6 summarizes configurations.

Table 6: **Run configurations.** TS = Tabula Sapiens. Pairs with evidence are a subset filtered to cells containing both source and target.

Run group	Dataset	Cells	Pairs
Kidney (4 seeds)	TS kidney	120	120+120
Lung	TS lung	120	120+120
Immune (2 seeds)	TS immune	120	120+120
Krasnow lung	Krasnow	40	30+30
Lung (fast)	TS lung	60	60+60
Lung (high-pair)	TS lung	40	160+160
Kidney (reduced)	TS kidney	300	80+80
Lung (reduced)	TS lung	240	80+80
Adamson (scaled)	Adamson	60	40+40
Adamson (large)	Adamson	60	100+100

Conclusion

Causal interventions on scGPT token representations enrich for curated TF–target edges, particularly in lung, and mechanistic tracing localizes components for specific pairs. The strongest result is lung tissue AUPR up to 0.78 against DoRothEA references with permutation p=0.001, a signal that survives across pair counts, robustness runs, and high-confidence reference subsets. Kidney enrichment is modest but reproducible across seeds, and immune tissue remains at baseline, together delineating where the model has and has not learned directional regulatory structure. A matched benchmark against eleven classical GRN inference methods spanning six algorithmic families confirms that scGPT leads the strongest classical baseline in all three tissues, with the lung gap substantially larger than the kidney or immune gaps, consistent with the tissue-level pattern of the main runs.

Several directions follow naturally from these findings. First, scaling runs to GPU-enabled full-tissue evaluations would increase pair counts by an order of magnitude, sharpening the statistical resolution for tissues where current enrichment is marginal. Second, replacing the mean-across-features readout with richer target representations, such as per-feature cosine similarity or learned probes, could recover regulatory effects that a single scalar compresses away. Third, applying the same causal intervention framework to other single-cell foundation models (Geneformer, scFoundation) would reveal whether tissue-dependent regulatory encoding is architecture-specific or a general property of expression-trained transformers. Finally, perturbation validation using tissue-matched Perturb-seq datasets, particularly from lung epithelial cells where model performance is strongest, would close the gap between representational causal effects and experimental ground truth that the Adamson K562 comparison currently leaves open.

More broadly, the pipeline we describe, combining token-level causal interventions with component-level activation patching, provides a general template for mechanistic audits of biological foundation models. As these models are increasingly deployed for cell-type annotation, perturbation prediction, and drug target prioritization, understanding which internal components encode which biological relationships becomes a practical safety and reliability concern, not merely an academic curiosity. The artifacts and evaluation framework accompanying this work are designed to lower the barrier for such audits across tissues, models, and regulatory contexts.

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Data availability statement

All primary datasets used in this study are publicly available. The Tabula Sapiens atlas [50] is available at `https://tabula-sapiens-portal.ds.czbiohub.org`. The Krasnow lung atlas [51] is available through the Human Lung Cell Atlas at `https://hlca.ds.czbiohub.org`. The Adamson CRISPR perturbation dataset [52, 60] is available on Harvard Dataverse at `https://doi.org/10.7910/DVN/BIKGRL`. TRRUST v2 [57] is available at `https://www.grnpedia.org/trrust`. DoRothEA regulons [58] are available through the decoupler Python package at `https://github.com/saezlab/decoupler-py`. The scGPT whole-human pretrained checkpoint [2] is available at `https://github.com/bowang-lab/scGPT`.

All source code for the causal intervention pipeline, evaluation scripts, run configurations, and figure generation code is available at `https://github.com/Biodyn-AI/casual-intervention`. Causal scores,

intermediate outputs, and reproducibility artifacts are included in the repository.

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Supporting information

S1 Table. Full causal metrics summary across all run configurations. Complete AUPR, AUROC, and permutation p-values for every run group, reference database, and intervention type.

S1 Data. Source code and evaluation artifacts. Available at `https://github.com/Biodyn-AI/casual-intervention`. Includes run configurations, evaluation scripts, causal scores, and figure generation code.